



## Review Article

## Rethinking insulin resistance in aging: A reserve-oriented clinical framework

Virginia Boccardi<sup>a,\*</sup>, Alan J. Sinclair<sup>b</sup><sup>a</sup> Division of Gerontology and Geriatrics, Department of Medicine and Surgery, University of Perugia, Perugia 06132, Italy<sup>b</sup> Foundation for Diabetes Research in Older People (FDROP) and King's College, London, UK

## ARTICLE INFO

## Keywords:

Ageing  
Insulin  
Resilience  
Stress

## ABSTRACT

Ageing represents one of the strongest non-modifiable determinants of insulin resistance (IR), a condition that extends well beyond impaired glucose handling and underling a broad spectrum of metabolic, cardiovascular, and neuropsychiatric disorders. In older adults, IR emerges from the progressive loss of physiological reserve across multiple organ systems rather than from isolated defects in insulin signalling. This narrative review examines the metabolic, inflammatory, and hormonal mechanisms linking ageing to insulin resistance, with a specific focus on skeletal muscle deterioration, adipose tissue remodelling, mitochondrial dysfunction, chronic low-grade inflammation, and cellular senescence. Age-related sarcopenia and myosteatosis compromise peripheral glucose disposal, while visceral adipose tissue expansion and adipocyte senescence promote a pro-inflammatory and insulin-desensitizing milieu. These peripheral alterations are amplified by inflammation, mitochondrial–endoplasmic reticulum dysfunction, and endocrine dysregulation involving growth hormone, sex steroids, and adipokines. Importantly, insulin resistance in ageing is increasingly recognized as a systemic condition affecting brain metabolism, thereby contributing to cognitive decline, depression, and frailty. Understanding insulin resistance as a multisystem failure of metabolic resilience provides a conceptual framework for integrated preventive and therapeutic strategies in older adults, combining lifestyle interventions, targeted pharmacological approaches, and emerging geroscience-based therapies.

## 1. Introduction: Insulin resistance as loss of metabolic resilience in ageing

Insulin resistance (IR) is conventionally framed as a metabolic defect leading to impaired glucose uptake and hyperglycaemia, primarily discussed in the context of obesity and type 2 diabetes mellitus (DeFronzo et al., 2015). However, this reductionist view becomes increasingly inadequate when applied to older adults. In ageing populations, insulin resistance frequently emerges in the absence of overt obesity, precedes the onset of diabetes by many years, and manifests together with functional decline, frailty, cognitive impairment, and mood disorders (Barzilai et al., 2012; Abbatecola, Paolisso, 2008). These observations suggest that, in later life, insulin resistance cannot be interpreted solely as a disorder of glucose metabolism, but rather as a systemic marker of declining physiological reserve (Ferrucci et al., 2018).

Ageing is characterized by substantial heterogeneity in metabolic trajectories. Individuals of similar chronological age often display

markedly divergent insulin sensitivity, body composition, and clinical outcomes (Clegg et al., 2013). This variability challenges the notion of insulin resistance as an inevitable consequence of ageing and instead points toward a complex interaction between biological ageing processes, tissue-specific vulnerability, and cumulative life-course exposures (Kirkwood, 2005). Thus, the term metabolic ageing is used to describe the progressive, age-associated decline in metabolic flexibility and regulatory efficiency (Khalaf et al., 2025). Within this framework, metabolic ageing includes impaired insulin signaling, reduced capacity to adapt glucose and lipid metabolism to physiological stressors, and a gradual loss of systemic resilience. In this context, insulin resistance appears less as a binary metabolic abnormality and more as a gradual failure of adaptive capacity across multiple organ systems (Fried et al., 2001).

Skeletal muscle, adipose tissue, and the central nervous system represent critical and interconnected nodes in this process. Age-related loss of muscle mass and quality reduces the primary site of insulin-

\* Correspondence to: Division of Gerontology and Geriatrics, Department of Medicine and Surgery, University of Perugia, Santa Maria della Misericordia Hospital, Piazzale Gambuli 1, Perugia 06132, Italy

E-mail address: [virginia.boccardi@unipg.it](mailto:virginia.boccardi@unipg.it) (V. Boccardi).

<https://doi.org/10.1016/j.arr.2026.103180>

Received 18 January 2026; Received in revised form 1 April 2026; Accepted 13 May 2026

Available online 17 May 2026

1568-1637/© 2026 The Author(s). Published by Elsevier B.V. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

mediated glucose disposal, while qualitative changes such as myosteatosis and anabolic resistance further impair metabolic responsiveness (Volpi et al., 2018; Goodpaster et al., 2001). Concurrently, adipose tissue undergoes a redistribution toward visceral depots, accompanied by immune infiltration, chronic inflammation, and cellular senescence (Xu et al., 2015). These changes transform adipose tissue from an energy storage organ into an active source of inflammatory and endocrine signals that amplify systemic insulin resistance (Ahmed et al., 2021; Hotamisligil, 2006).

Importantly, growing evidence indicates that insulin resistance in ageing extends beyond peripheral tissues. The brain, a highly insulin-sensitive organ, exhibits age-related alterations in insulin signalling that are increasingly implicated in cognitive decline and neurodegeneration (Milstein, Ferris, 2021; Craft, 2005). Reduced cerebral glucose utilization, neuroinflammation, and synaptic dysfunction associated with brain insulin resistance provide a mechanistic link between metabolic dysregulation and late-life cognitive and affective disorders (Arnold et al., 2018; De la Monte, 2017). This central component challenges the traditional peripheral-centric view of insulin resistance and underscores its role as a multisystem condition.

At the biological level, these tissue-specific alterations are unified by fundamental ageing mechanisms, including chronic low-grade inflammation (inflammageing), mitochondrial dysfunction, impaired organellar crosstalk, genomic instability, a decline in stem cell capacity, and the accumulation of senescent cells with a pro-inflammatory secretory phenotype (Franceschi et al., 2018; López-Otín et al., 2013; Zhang et al., 2025; Liu et al., 2018). Hormonal changes involving growth hormone, sex steroids, and adipokines further modulate these processes, shaping body composition, metabolic flexibility, and insulin sensitivity across the ageing trajectory (Veldhuis et al., 2011).

In this review, we adopt an integrative and translational narrative approach to examine insulin resistance in older adults within the context of biological ageing. Rather than considering insulin resistance solely as a defect in insulin signalling, we explore evidence suggesting that it may reflect a progressive decline in physiological resilience—defined as the capacity of biological systems to adapt to stressors, preserve functional integrity, and recover toward baseline following perturbations (Ferrucci et al., 2018; Weavers, 2024). By synthesizing metabolic, inflammatory, hormonal, and neurobiological pathways, this review seeks to delineate how insulin resistance intersects with key ageing-related phenotypes, including frailty, cognitive decline, and multimorbidity, and to highlight implications for more targeted and individualized strategies for prevention and intervention in ageing populations (Cesari et al., 2014).

## 2. Skeletal muscle ageing as the primary determinant of insulin resistance in later life

Skeletal muscle represents the dominant site of insulin-mediated glucose disposal, accounting for the majority of postprandial glucose uptake, and therefore constitutes the principal functional buffer against glycaemic excursions (DeFronzo et al., 2015). In younger individuals, this buffering capacity is sustained by preserved muscle mass, mitochondrial oxidative capacity, and intact anabolic signalling. With ageing, however, skeletal muscle progressively loses not only quantity but, more importantly, qualitative metabolic competence, transforming from a site of metabolic flexibility into a critical vulnerability node for insulin resistance (Larsson et al., 2019).

Although age-related muscle decline is commonly subsumed under the construct of sarcopenia, classically defined by reductions in muscle mass and strength, histological, imaging, and interventional studies indicate that ageing muscle undergoes earlier qualitative and metabolic alterations (Nowakowska-Lipiec et al., 2025). These include a progressive loss of muscle fibers, preferential atrophy and loss of fast-twitch (type II) fibers, increased intramuscular fat infiltration, and impaired mitochondrial and insulin-mediated metabolism, which precede and likely contribute to overt muscle atrophy and functional decline

(Sinclair et al., 2025). Controlled metabolic trials using hyperinsulinemic–euglycemic clamps combined with muscle biopsies have demonstrated that older adults exhibit markedly reduced insulin-stimulated glucose uptake despite relatively preserved muscle mass, implicating intrinsic defects in muscle insulin signalling and substrate oxidation rather than simple loss of tissue quantity (Shulman, 2000; Petersen et al., 2003). Again, recent work has emphasized that frailty in older adults with diabetes is associated with unfavorable body composition changes, including sarcopenic fat accumulation and metabolic inefficiency, which may modify both insulin sensitivity and therapeutic responses (Sinclair et al., 2025). These observations reinforce the view of frailty as a metabolically active condition rather than a purely functional phenotype.

Randomized and mechanistic trials further demonstrate that qualitative alterations in muscle composition—particularly intramyocellular lipid accumulation and mitochondrial dysfunction—are central to this process (Tero-Vescan et al., 2025; Scalabrín et al., 2025). Studies employing magnetic resonance spectroscopy and biopsy-based lipid quantification show that increased intramyocellular lipid content is strongly associated with insulin resistance independent of total adiposity or body weight, even in lean and physically active individuals (Perseghin et al., 1999; Petersen et al., 2004). These lipid-related alterations impair insulin signalling through the accumulation of bioactive lipid intermediates, such as diacylglycerols and ceramides. Although earlier models emphasized direct inhibition of insulin receptor substrate proteins, accumulating evidence indicates that lipid-induced insulin resistance is mediated by a network of mechanisms, including activation of specific protein kinase C isoforms, disruption of membrane signaling domains, and mitochondrial perturbations. Notably, the extent to which insulin receptor substrate (IRS) dysfunction represents a primary driver of insulin resistance in human skeletal muscle remains controversial, suggesting that defects in insulin action may arise from multiple, partially independent pathways (James et al., 2021; Fazakerley et al., 2019).

From a temporal perspective, longitudinal and familial studies indicate that skeletal muscle insulin resistance represents an early and initiating defect in the trajectory toward type 2 diabetes (DeFronzo, Tripathy, 2009). In offspring of patients with type 2 diabetes, clamp studies reveal impaired muscle glucose uptake and reduced mitochondrial oxidative phosphorylation decades before the development of hyperglycaemia or  $\beta$ -cell dysfunction (Petersen et al., 2004). This pre-clinical phase is characterized by reduced insulin-stimulated glycogen synthesis and diminished mitochondrial ATP production (Petersen et al., 2004). Collectively, these findings support the view that skeletal muscle may represent an early site of metabolic dysregulation rather than merely a secondary target of systemic disease. Anabolic resistance further amplifies this vulnerability. Randomized controlled trials have demonstrated that older adults require substantially higher protein doses to stimulate muscle protein synthesis compared with younger individuals, yet even supraphysiological amino acid provision fails to fully restore anabolic responsiveness in the presence of insulin resistance (Volpi et al., 2000; Cuthbertson et al., 2005). Similarly, resistance exercise trials suggest attenuated hypertrophic and metabolic responses in insulin-resistant older adults (Lin et al., 2026), supporting a bidirectional relationship in which insulin resistance may contribute to muscle loss, while declining muscle quality may further worsen insulin sensitivity. Experimental models of short-term physical inactivity provide compelling evidence for the fragility of muscle metabolic reserve in ageing. Bed-rest and step-reduction trials demonstrate that even brief periods of immobilization induce rapid declines in insulin sensitivity, mitochondrial function, and muscle protein synthesis, with older adults exhibiting disproportionately greater metabolic deterioration compared with younger counterparts (Breen et al., 2013; Dirks et al., 2016). These findings offer a mechanistic explanation for the marked acceleration of insulin resistance observed during hospitalization and acute illness in older populations.

Beyond its role in glucose disposal, skeletal muscle functions as an endocrine organ through the secretion of myokines that modulate systemic insulin sensitivity and inflammation (Merz, Thurmond, 2020). Exercise intervention trials consistently show that restoration of muscle contractile activity improves whole-body insulin sensitivity partly through favorable shifts in myokine signalling, including reductions in pro-inflammatory (Gonzalez-Gil, Elizondo-Montemayor, 2020) mediators and enhancement of insulin-sensitizing pathways (Pedersen and Febbraio, 2012). Ageing and inactivity blunt these adaptive endocrine responses, further impairing inter-organ metabolic communication.

Clinically, these trial-based observations challenge adiposity-centered models of metabolic risk. Intervention studies reveal that improvements in insulin sensitivity achieved through resistance training or combined exercise–nutrition strategies correlate more strongly with gains in muscle quality and mitochondrial function than with changes in body weight or fat mass (Short et al., 2003; Stephens, Sparks, 2007). Consequently, older adults with normal body mass index but reduced muscle metabolic reserve may exhibit profound insulin resistance, while individuals with higher body weight but preserved muscle function may remain metabolically resilient (Damanti et al., 2024).

Taken together, this evidence supports the concept that skeletal muscle ageing is a foundational determinant of insulin resistance in later life. Rather than representing parallel consequences of ageing, sarcopenia and insulin resistance emerge as tightly coupled processes driven by progressive erosion of muscle metabolic reserve. This framework reframes insulin resistance as an early manifestation of functional decline and provides a strong rationale for preventive and therapeutic strategies that prioritize preservation of muscle quality and metabolic capacity in ageing populations.

### 3. Adipose tissue ageing: From energy storage to endocrine amplifier of insulin resistance

In the ageing organism, adipose tissue undergoes a profound functional reprogramming that extends well beyond quantitative changes in fat mass. While increased adiposity—particularly in visceral depots—has long been associated with insulin resistance, longitudinal human studies demonstrate that ageing per se induces qualitative alterations in adipose tissue biology that are only weakly correlated with body mass index or total fat mass (Tchkonina et al., 2010; Stout et al., 2017). In older adults, adipose tissue progressively shifts from a relatively inert energy reservoir to an active endocrine and inflammatory organ that amplifies systemic insulin resistance.

One of the earliest and most consistent age-related changes is the redistribution of fat from subcutaneous to visceral compartments. Imaging-based cohort studies using CT and MRI show that visceral fat accumulation increases with age even in weight-stable individuals and in the absence of overt obesity (Kuk et al., 2009; Porter et al., 2023). Visceral adipose tissue is metabolically distinct, characterized by higher basal lipolytic activity and increased release of free fatty acids into the portal circulation, thereby promoting hepatic insulin resistance, dyslipidaemia, and ectopic lipid deposition (Bergman et al., 2007; Item and Konrad, 2012). These alterations establish a central liver–visceral fat axis of metabolic dysfunction that becomes increasingly prominent with ageing.

However, fat redistribution alone does not fully account for age-related insulin resistance. Human biopsy studies and translational trials demonstrate that ageing adipose tissue undergoes marked immune remodelling, characterized by increased infiltration of pro-inflammatory macrophages, a shift toward M1-like phenotypes, altered T-cell populations, and sustained low-grade inflammation (Xu et al., 2015; Ferrucci and Fabbri, 2018). Importantly, this inflammatory phenotype is observed in older adults with normal or mildly elevated body mass index (BMI), indicating that adipose tissue inflammation reflects tissue dysfunction rather than adiposity burden alone (Fabbri et al., 2016).

Cellular senescence has emerged as a central driver of adipose tissue dysfunction in ageing. Studies in humans combining adipose biopsies with senescence markers demonstrate an age-dependent accumulation of senescent adipocytes and stromal cells, which secrete a senescence-associated secretory phenotype (SASP) enriched in pro-inflammatory cytokines, chemokines, and matrix-remodeling enzymes (Palmer et al., 2019). Experimental senolytic interventions in older adults provide proof-of-concept that reducing senescent cell burden improves adipose tissue function and systemic insulin sensitivity, directly implicating senescence as a causal contributor rather than a passive correlate of metabolic ageing (Justice et al., 2019; Hickson et al., 2019).

This senescent burden impairs adipose tissue expandability and lipid storage capacity, promoting spillover of lipids toward ectopic depots such as liver and skeletal muscle. Human metabolic studies demonstrate that limited adipose expandability is associated with increased ectopic fat accumulation and insulin resistance independently of total fat mass, reframing lipotoxicity as a failure of adipose buffering capacity rather than simple fat excess (Virtue and Vidal-Puig, 2010; Ladeiras-Lopes et al., 2024).

Adipokine dysregulation further amplifies metabolic impairment. Ageing is consistently associated with reduced circulating adiponectin levels and altered leptin signaling, both of which play key roles in insulin sensitivity and energy homeostasis (Cnop et al., 2003; Kizer et al., 2011). Declining adiponectin reduces fatty acid oxidation and enhances hepatic gluconeogenesis, while leptin resistance—particularly prevalent in older adults with visceral adiposity—impairs central appetite regulation and exacerbates metabolic inflexibility (Myers et al., 2010). These endocrine changes reinforce the systemic impact of adipose tissue ageing on insulin resistance.

The interaction between adipose tissue and skeletal muscle is particularly relevant in later life. Stable isotope and clamp studies demonstrate that elevated circulating free fatty acids and pro-inflammatory adipokines derived from dysfunctional adipose tissue directly impair muscle insulin signaling and promote intramyocellular lipid accumulation (Boden, 2008; Petersen and Shulman, 2018). Conversely, reduced muscle mass and physical inactivity limit lipid oxidation capacity, further increasing lipid burden on adipose tissue. This bidirectional crosstalk promotes the development of sarcopenic obesity, a phenotype strongly associated with frailty, disability, and adverse clinical outcomes in older adults (Zamboni et al., 2008; Atkins et al., 2014).

Clinically, these data challenge traditional adiposity-centered models of metabolic risk. Prospective studies show that BMI poorly predicts insulin resistance and cardiometabolic outcomes in older adults when compared with measures of visceral fat, ectopic fat, and inflammatory biomarkers (Kuk et al., 2009; Britton et al., 2013). Older individuals with normal body weight may harbor profoundly dysfunctional adipose tissue, while others with higher adiposity but preserved adipose function remain metabolically stable. This dissociation underscores the need for markers that reflect adipose tissue biology rather than quantity alone.

Within this framework, ageing adipose tissue emerges as an endocrine amplifier of insulin resistance, translating local cellular senescence and immune dysregulation into systemic metabolic consequences. Rather than serving merely as a passive fat depot, adipose tissue acts as a central regulator of metabolic resilience in ageing. Targeting adipose tissue inflammation, senescence, and endocrine dysfunction therefore represents a critical therapeutic leverage point for mitigating insulin resistance and its downstream clinical sequelae in older adults. These processes do not act independently but form a self-reinforcing biological network that progressively erodes metabolic resilience.

### 4. Inflammageing, cellular senescence, and mitochondrial dysfunction as a unifying mechanistic axis

The metabolic alterations observed in ageing skeletal muscle and

adipose tissue do not arise in isolation but reflect the convergence of systemic biological processes that progressively erode metabolic resilience. Among these, chronic low-grade inflammation, cellular senescence, and mitochondrial dysfunction constitute a tightly interconnected mechanistic axis that links tissue-specific insulin resistance to whole-body metabolic decline (López-Otín et al., 2013; Ferrucci and Fabbri, 2018).

Inflammageing describes a persistent state of subclinical inflammation that emerges with advancing age, even in the absence of overt infection or autoimmune disease. Large longitudinal cohort studies have consistently demonstrated age-associated elevations in circulating inflammatory mediators such as tumor necrosis factor- $\alpha$ , interleukin-6, and interleukin-1 $\beta$ , which independently predict insulin resistance, mobility decline, and mortality in older adults (Franceschi et al., 2018; Ferrucci et al., 2018). Although modest in magnitude, the chronic activation of these cytokines exerts profound effects on insulin signaling through stress-activated pathways. Experimental and translational studies show that activation of c-Jun N-terminal kinase (JNK) and I $\kappa$ B kinase  $\beta$  (IKK $\beta$ ) interferes with insulin receptor substrate phosphorylation, suppresses PI3K–Akt signalling, and reduces GLUT4 translocation across skeletal muscle, adipose tissue, and liver (Hotamisligil, 2006; Petersen and Shulman, 2018). Importantly, these inflammatory effects on insulin signalling occur independently of adiposity, underscoring inflammation as a primary driver rather than a secondary correlate of metabolic ageing.

Crucially, inflammageing should not be interpreted as a passive epiphenomenon of ageing. Its origins are multifactorial and include lifelong antigenic burden, immunosenescence, altered gut permeability, and the accumulation of damaged macromolecules and organelles (Franceschi et al., 2018). In this context, adipose tissue and skeletal muscle—particularly when affected by lipid infiltration and senescence—become active sources of inflammatory mediators, reinforcing systemic inflammation and perpetuating insulin resistance through feed-forward loops (Fabbri et al., 2016; Stout et al., 2017).

Cellular senescence provides a direct mechanistic bridge between tissue ageing and inflammageing. Senescent cells accumulate with age in multiple organs, including adipose tissue, skeletal muscle, and the vascular endothelium (Palmer et al., 2019). While senescence initially serves as a protective response to cellular stress, its chronic persistence becomes maladaptive. Senescent cells adopt a senescence-associated secretory phenotype (SASP), characterized by the release of pro-inflammatory cytokines, chemokines, growth factors, and matrix-remodelling enzymes that disrupt tissue architecture and impair insulin signaling in neighboring cells.

In adipose tissue, senescent adipocytes and stromal progenitors reduce tissue expandability, promote fibrosis, and impair lipid storage capacity, thereby favoring ectopic fat deposition in liver and skeletal muscle (Tchkonia et al., 2010; Palmer et al., 2019). In skeletal muscle, senescence limits satellite cell function, blunts regenerative capacity, and contributes to anabolic resistance, exacerbating muscle loss and metabolic dysfunction. Importantly, SASP factors exert endocrine-like effects, allowing localized senescent burden to translate into systemic insulin resistance. Proof-of-concept human trials using senolytic agents demonstrate that reducing senescent cell burden improves insulin sensitivity and physical function, supporting a causal role for senescence in metabolic ageing (Justice et al., 2019; Hickson et al., 2019).

Mitochondrial dysfunction further integrates these processes. Ageing is associated with reduced mitochondrial biogenesis, impaired oxidative phosphorylation, and defective mitophagy, leading to the accumulation of dysfunctional mitochondria in insulin-sensitive tissues (Petersen et al., 2003). These alterations limit metabolic flexibility, promote lipid accumulation, and increase reactive oxygen species production. Oxidative stress, in turn, activates inflammatory signaling cascades and accelerates cellular senescence, establishing a self-reinforcing cycle linking mitochondrial dysfunction, inflammation, and insulin resistance (Picard et al., 2018).

**Table 1**

Biological drivers of insulin resistance in ageing and their tissue-specific effects.

| Biological process                    | Primary tissues involved                      | Key mechanisms                                                                                                          | Metabolic consequences                                                           |
|---------------------------------------|-----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| Inflammageing                         | Skeletal muscle, adipose tissue, liver, brain | Chronic activation of TNF- $\alpha$ , IL-6, IL-1 $\beta$ ; JNK and IKK $\beta$ signaling; impaired IRS–PI3K–Akt pathway | Systemic insulin resistance, reduced glucose uptake, impaired anabolic signaling |
| Cellular senescence                   | Adipose tissue, skeletal muscle, endothelium  | Accumulation of senescent cells; SASP secretion; fibrosis and impaired tissue regeneration                              | Loss of adipose expandability, anabolic resistance, ectopic fat deposition       |
| Mitochondrial dysfunction             | Muscle, adipose tissue, brain                 | Reduced oxidative phosphorylation; impaired mitophagy; increased ROS                                                    | Metabolic inflexibility, lipid accumulation, energy deficit                      |
| Organelle crosstalk disruption (MAMs) | Muscle, liver, adipose tissue                 | Altered ER–mitochondria calcium and lipid signaling                                                                     | Impaired insulin signaling across multiple organs                                |

Akt, protein kinase B; ER, endoplasmic reticulum; IKK $\beta$ , inhibitor of nuclear factor kappa-B kinase subunit beta; IL-1 $\beta$ , interleukin-1 beta; IL-6, interleukin-6; IRS, insulin receptor substrate; JNK, c-Jun N-terminal kinase; MAMs, mitochondria-associated endoplasmic reticulum membranes; PI3K, phosphoinositide 3-kinase; ROS, reactive oxygen species; SASP, senescence-associated secretory phenotype; TNF- $\alpha$ , tumor necrosis factor alpha.

Disruption of mitochondrial–endoplasmic reticulum communication represents a critical and increasingly recognized mechanism in this cycle. Mitochondria-associated endoplasmic reticulum membranes (MAMs) coordinate calcium flux, lipid metabolism, and insulin signaling. Human and experimental studies demonstrate that age-related alterations in MAM structure impair calcium homeostasis, exacerbate oxidative stress, and blunt insulin signaling in muscle, adipose tissue, and liver (Arruda et al., 2014). This organelle-level dysfunction provides a mechanistic explanation for the systemic nature of insulin resistance in ageing, transcending single-tissue abnormalities.

Together, inflammageing, cellular senescence, and mitochondrial dysfunction form an integrated biological axis that drives insulin resistance as a hallmark of ageing rather than a discrete metabolic disorder (López-Otín et al., 2013; Ferrucci and Fabbri, 2018). These processes mutually reinforce one another, converting localized tissue damage into systemic metabolic impairment. This framework helps explain why insulin resistance in older adults often persists despite modest weight loss or glycaemic control and why glucose-centric interventions frequently fail to modify the broader clinical phenotype (Table 1).

## 5. Brain insulin resistance: Linking metabolic ageing to cognitive decline and mood disorders

The systemic nature of insulin resistance becomes particularly evident when its effects on the central nervous system are considered. For decades, insulin resistance was conceptualized primarily as a peripheral metabolic disorder. However, this peripheral-centric model fails to explain the strong and consistent association between insulin resistance and late-life cognitive decline, depression, and neurodegenerative vulnerability. Converging evidence now demonstrates that insulin resistance extends to the central nervous system, where impaired insulin signaling contributes directly to brain ageing and dysfunction (Arnold et al., 2018).

The brain is a highly energy-demanding organ that relies predominantly on glucose metabolism. Although neuronal glucose uptake is partially insulin-independent, insulin signaling in the brain regulates synaptic plasticity, neurotransmitter turnover, neuronal survival, and



neuroinflammatory responses (Craft, 2005; Arnold et al., 2018). Insulin receptors are abundantly expressed in regions critical for cognition and mood regulation, including the hippocampus, prefrontal cortex, and hypothalamus. Ageing is associated with reduced insulin transport across the blood–brain barrier and impaired neuronal insulin receptor signaling, leading to brain insulin resistance (Banks et al., 2012).

Neuroimaging and metabolic studies demonstrate that peripheral insulin resistance is strongly associated with reduced cerebral glucose metabolism, even in non-diabetic older adults (Willette et al., 2013). This cerebral hypometabolism precedes overt cognitive impairment and mirrors patterns observed in prodromal Alzheimer's disease, suggesting that metabolic dysfunction is an upstream driver of neurodegeneration rather than a secondary consequence (Raut et al., 2023). Brain insulin resistance contributes to neuronal energy deficits, synaptic dysfunction, and impaired network connectivity, all of which are central features of cognitive ageing.

Neuroinflammation represents a critical mediator linking systemic insulin resistance to brain dysfunction. Inflammageing extends to the central nervous system via circulating cytokines, microglial priming, and age-related disruption of the blood–brain barrier (Perry and Holmes, 2014). Insulin-resistant states promote microglial activation and pro-inflammatory signaling, which further interferes with neuronal insulin signaling and exacerbates synaptic dysfunction, creating a vicious cycle of metabolic and inflammatory stress (Arnold et al., 2018).

Beyond cognition, brain insulin resistance is increasingly implicated in mood disorders, particularly late-life depression. Insulin signaling modulates serotonergic and dopaminergic neurotransmission, neuroplasticity, and stress responsiveness. Epidemiological and mechanistic studies demonstrate that insulin resistance is associated with depressive symptoms independently of vascular disease or overt diabetes (Lyra e Silva et al., 2019). These findings support a direct neurobiological role for insulin resistance in affective regulation.

Brain insulin resistance should therefore be interpreted not as an isolated neurological phenomenon but as the downstream expression of a systemic failure of metabolic resilience. Skeletal muscle insulin resistance limits peripheral glucose disposal, adipose tissue dysfunction amplifies inflammatory signaling, and mitochondrial impairment reduces metabolic flexibility (Popescu et al., 2026). Together, these processes increase metabolic stress on the ageing brain, which is particularly vulnerable to energy deficits. This convergence provides a mechanistic explanation for the frequent co-occurrence of insulin resistance, cognitive decline, frailty, and mood disorders in older adults.

Recognizing brain insulin resistance as a core component of metabolic ageing challenges traditional disease boundaries and supports a unified biological continuum linking metabolic, cognitive, and affective disorders. Clinically, this perspective underscores the limitations of glucose-centric management strategies and supports earlier, multidimensional interventions targeting inflammation, metabolic flexibility, and insulin signaling before irreversible neurodegeneration occurs. Emerging approaches, including lifestyle modulation and experimental strategies such as intranasal insulin delivery, derive their rationale directly from this integrated framework (Craft et al., 2017).

## 6. Therapeutic implications and future perspectives: Targeting metabolic resilience rather than glycaemia

Reframing insulin resistance in ageing as a multisystem failure of metabolic resilience has direct implications for prevention and treatment. In older adults, strategies focused exclusively on glycemic control or weight reduction often fail to modify the broader clinical trajectory, particularly with respect to functional decline, frailty, and cognitive outcomes. An effective therapeutic approach must therefore address the upstream biological processes that erode physiological reserve across skeletal muscle, adipose tissue, and brain. Interestingly, evidence from longitudinal studies in non-diabetic older adults suggests that the relationship between insulin resistance and adverse outcomes is not linear.

**Table 2**

Therapeutic strategies targeting components of metabolic reserve in ageing.

| Intervention                   | Primary biological target           | Mechanism of action                                                  | Impact on metabolic reserve                                                          |
|--------------------------------|-------------------------------------|----------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| Resistance training            | Skeletal muscle                     | ↑ Insulin-stimulated glucose uptake; ↑ muscle protein synthesis      | ↑ Muscle mass, strength, and functional reserve                                      |
| Aerobic exercise               | Mitochondria                        | ↑ Oxidative capacity; ↑ lipid oxidation                              | ↑ Metabolic flexibility and endurance                                                |
| Adequate protein intake        | Muscle anabolism                    | Overcomes anabolic resistance                                        | Preservation of muscle mass and function                                             |
| Dietary quality / fiber intake | Gut microbiota                      | ↑ SCFA production; ↓ proteolytic fermentation                        | ↓ Inflammation; improved metabolic signaling                                         |
| GLP-1 receptor agonists        | Energy balance, appetite regulation | ↓ Energy intake; modulation of inflammatory pathways                 | Potential improvement in metabolic regulation (risk of lean mass loss if unbalanced) |
| SGLT2 inhibitors               | Energy metabolism, nutrient sensing | ↑ Glycosuria; caloric loss; metabolic shift toward lipid utilization | CR-like metabolic adaptation; potential systemic benefits                            |
| Bariatric surgery              | Whole-body metabolism               | Profound metabolic remodeling                                        | Potential diabetes remission; long-term metabolic reset                              |
| Senotherapeutics               | Cellular senescence                 | ↓ Senescent cell burden; ↓ SASP                                      | Experimental improvement of systemic resilience                                      |
| Intranasal insulin             | Brain insulin signaling             | Direct CNS insulin modulation                                        | Potential improvement in cognitive reserve                                           |

CR, caloric restriction; GLP-1 RA, glucagon-like peptide-1 receptor agonists; IR, insulin resistance; SGLT2i, sodium–glucose cotransporter-2 inhibitors; SCFA, short-chain fatty acids; CNS, central nervous system; SASP, senescence-associated secretory phenotype.

Moderate insulin resistance has been associated with preserved physical function and lower mortality, whereas more pronounced insulin resistance, particularly when accompanied by frailty, inflammation, or adverse body composition changes, is linked to functional decline and increased mortality (Rodríguez-Mañas et al., 2022). These findings support the concept of insulin resistance as a context-dependent phenomenon rather than a uniformly maladaptive state in ageing. Table 2 summarizes major interventions according to their primary biological targets, mechanisms of action, and their potential impact on metabolic reserve. Rather than focusing solely on glycaemic control, this framework highlights how different strategies act on complementary pathways—including skeletal muscle, mitochondrial function, gut microbiota, inflammation, and brain insulin signaling—to influence insulin sensitivity and systemic resilience. Lifestyle interventions remain the cornerstone of therapy, while pharmacological and emerging geroscience-based approaches may provide additional benefits by targeting upstream mechanisms of metabolic dysfunction. The overall model supports a shift toward an integrated, reserve-oriented therapeutic paradigm in ageing populations.

### 6.1. Lifestyle interventions as foundational modulators of metabolic reserve

Lifestyle interventions remain the most robust and immediately translatable strategy for mitigating insulin resistance in ageing (Knowler et al., 2002a, 2002b; Hawley and Lessard, 2008). However, their effectiveness depends on appropriate targeting. In older adults, physical activity should be conceptualized not merely to improve insulin

sensitivity, but as a primary intervention to preserve muscle metabolic reserve. Randomized controlled trials demonstrate that resistance training, alone or combined with aerobic exercise, directly counteracts sarcopenia, improves mitochondrial oxidative capacity, and enhances insulin-stimulated glucose uptake (Short et al., 2003; Holten et al., 2004). Importantly, resistance exercise improves insulin sensitivity even in the absence of significant weight loss, reinforcing the primacy of muscle quality over adiposity reduction (Ibanez et al., 2005; Stephens, Sparks, 2007). The European MID-FRAIL study demonstrated that a structured multimodal intervention centered on progressive resistance training significantly improved physical function, muscle strength, and frailty status in older adults with type 2 diabetes (Rodríguez-Mañas et al., 2019). Notably, these functional gains occurred in parallel with improvements in body composition and metabolic health, supporting the concept that targeted exercise interventions can partially reverse frailty-related features and restore physiological resilience even in the presence of chronic metabolic disease.

Nutritional strategies must similarly move beyond calorie-centric paradigms. Controlled feeding studies and intervention trials show that adequate protein intake is essential to overcome anabolic resistance and support muscle maintenance in older adults (Volpi et al., 2000; Cuthbertson et al., 2005). Moreover, dietary patterns characterized by higher protein quality, anti-inflammatory components, and lower glycaemic load are associated with improved insulin sensitivity and reduced systemic inflammation (Esposito et al., 2009; Estruch et al., 2018). In older adults, the timing, distribution, and per-meal protein dose appear to be as important as total caloric intake for preserving muscle mass and metabolic function (Mamerow et al., 2014). Collectively, these interventions represent low-risk, high-impact tools capable of modulating multiple components of the muscle–adipose–brain axis simultaneously.

## 6.2. Pharmacological approaches: from glucose lowering to disease modification

Increasing attention is now directed toward early disease-modifying strategies aimed at altering the trajectory of metabolic dysfunction rather than solely controlling glycaemia. This perspective is increasingly shared across diabetology, cardiovascular medicine, and geroscience, where targeting upstream mechanisms of disease progression has become a central objective (Cosentino et al., 2020; Davies et al., 2018).

Beyond glycaemic control, metabolic disease can be more appropriately interpreted as a multi-system condition involving interconnected pathways such as low-grade inflammation, nutrient sensing, and gut microbiota-derived signaling. The gut microbiome has emerged as a key modulator of host metabolism, contributing to insulin resistance and type 2 diabetes through the production of bioactive metabolites. Short-chain fatty acids derived from saccharolytic fermentation may exert beneficial metabolic effects, whereas proteolytic fermentation generates compounds potentially detrimental to metabolic and vascular health, underscoring the importance of dietary patterns in shaping metabolic trajectories (Canfora et al., 2019).

These processes are closely intertwined with chronic low-grade inflammation, which represents a central driver of metabolic dysfunction and its cardiovascular complications. Inflammatory pathways are activated early during type 2 diabetes and may persist despite standard metabolic control, contributing to a residual risk that is not fully addressed by conventional therapies. This supports the view of type 2 diabetes as a dynamic inflammatory trajectory, potentially amenable to modulation through early and targeted interventions (Pellegrini et al., 2024).

Rather than focusing exclusively on glycaemic lowering, emerging strategies aim to intervene on upstream mechanisms of disease progression, including insulin resistance, inflammation, and metabolic remodeling (Mosenzon et al., 2021). In this context, also bariatric surgery provides robust proof-of-concept that early and sustained

metabolic modulation can lead to durable improvements in glycaemic control and, in some cases, remission of type 2 diabetes (Look AHEAD Research Group, 2013; Knowler et al., 2002a, 2002b; Li et al., 2008; Courcoulas et al., 2024).

Pharmacological agents such as GLP-1 receptor agonists and SGLT2 inhibitors are increasingly investigated within this paradigm. GLP-1 receptor agonists, through appetite suppression, may counteract chronic overnutrition, a key driver of metabolic dysfunction and ageing-related processes, including systemic inflammation (Pellegrini et al., 2024). Similarly, SGLT2 inhibitors induce a net loss of calories and promote metabolic adaptations that partially resemble those observed during caloric restriction, including shifts in substrate utilization and modulation of nutrient-sensing pathways (La Grotta et al., 2022). These mechanisms may contribute to broader systemic effects extending beyond glycaemic control, although their ability to modify long-term disease trajectories remains to be fully established.

In older adults, however, these potential benefits must be balanced against important limitations. Subgroup analyses and geriatric-focused studies highlight risks including excessive weight loss, loss of lean mass, dehydration, and adverse interactions related to polypharmacy (Batsis et al., 2021; Morley, 2022). This is particularly relevant in frail individuals and in the context of sarcopenic obesity, where excess adiposity coexists with impaired muscle quality and reduced functional reserve (Sinclair et al., 2025). In such settings, pharmacologically induced weight loss does not necessarily translate into restoration of physiological reserve. Body composition studies indicate that weight reduction achieved without concurrent resistance training and adequate protein intake may exacerbate sarcopenia and functional decline (Villareal et al., 2011; Kim et al., 2022). These observations support a personalized, muscle-preserving therapeutic approach in older adults, in which pharmacological treatments are integrated with nutritional and exercise strategies rather than applied in isolation.

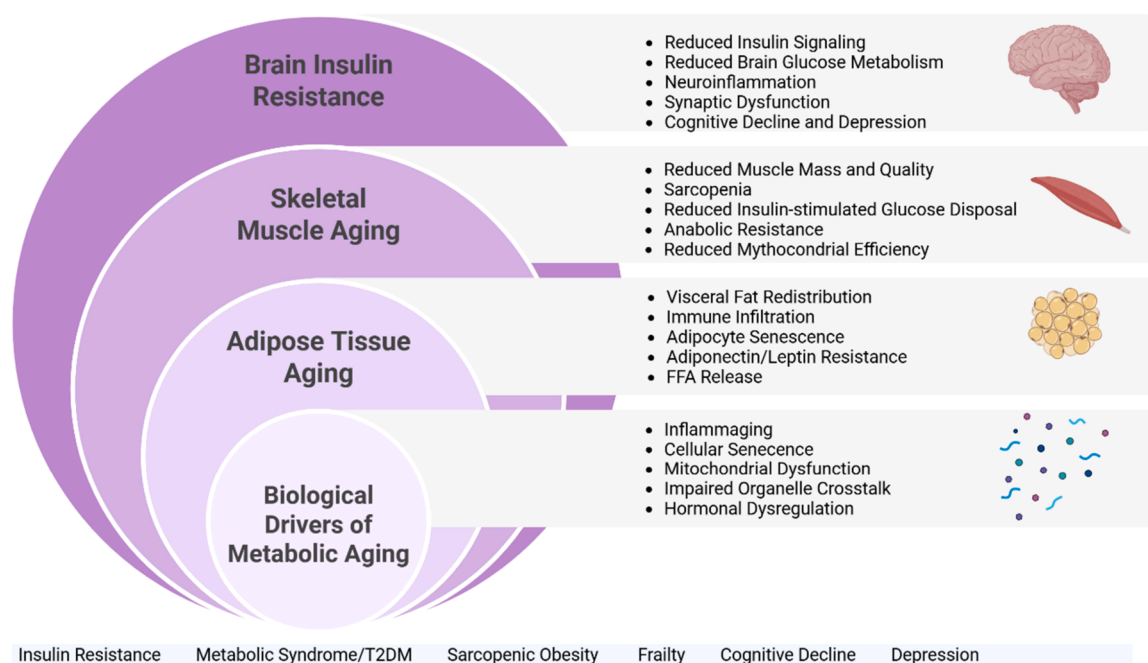
Importantly, a resilience-oriented framework should not obscure the continuing relevance of glycaemic control as an evidence-based therapeutic target. In individuals with recently diagnosed type 2 diabetes, particularly in those with shorter disease duration and low cardiovascular burden, early achievement of good glycaemic control has been associated with sustained reductions in both microvascular and macrovascular complications, supporting the concept that early metabolic exposure may influence long-term disease trajectories (Prattichizzo et al., 2020).

Landmark studies, including the UK Prospective Diabetes Study (UKPDS) and its extended long-term follow-up, have demonstrated that early intensive glycaemic control confers durable benefits over decades, including reductions in all-cause mortality, myocardial infarction, and microvascular complications, consistent with a persistent “legacy effect” (Holman et al., 2008; Adler et al., 2024). These findings suggest that achieving near-normoglycaemia soon after diagnosis may play a critical role in minimizing lifetime risk.

Within this integrated framework, the early use of disease-modifying pharmacological agents appears conceptually aligned with, rather than alternative to, glucose-centered care. Real-world evidence indicates that early introduction of sodium-glucose cotransporter-2 inhibitors may attenuate the increased cardiovascular risk associated with suboptimal glycaemic control in the initial phases of type 2 diabetes, suggesting a potential role in modifying early disease trajectories (Ceriello et al., 2023).

Mechanistic studies further support the biological plausibility that pharmacological interventions may influence pathways related to oxidative stress, endothelial dysfunction, and metabolic memory, although direct clinical evidence in this context remains limited (Pujadas et al., 2017).

Taken together, these observations support a unified therapeutic perspective in which early glycaemic control and early pharmacological intervention are interpreted as complementary strategies grounded in a shared principle: intervening early along the trajectory of metabolic



**Fig. 1.** Insulin resistance as a multisystem failure of metabolic resilience in ageing.

Ageing-related insulin resistance reflects a progressive loss of metabolic reserve across skeletal muscle, adipose tissue, and brain. Muscle ageing impairs insulin-mediated glucose disposal via sarcopenia, myosteatosis, and anabolic resistance, while visceral fat redistribution, immune remodeling, and adipose senescence amplify systemic insulin resistance. In parallel, brain insulin resistance is linked to cerebral hypometabolism, neuroinflammation, cognitive decline, and mood disorders. These changes converge on core ageing mechanisms— inflammaging, cellular senescence/SASP, mitochondrial dysfunction, disrupted mitochondria–endoplasmic reticulum crosstalk, and hormonal dysregulation—supporting insulin resistance as a hallmark of metabolic ageing rather than an isolated abnormality.

disease to influence its progression. Within this framework, glucose lowering remains essential, but is embedded within a broader approach aimed at preserving muscle function, limiting treatment-related vulnerability, delaying disease progression, and promoting healthy ageing. This integrated perspective aligns with emerging views in diabetology and geroscience, supporting early, multidimensional intervention as a strategy to modify disease trajectory rather than simply manage late-stage metabolic consequences.

### 6.3. Emerging geroscience-based strategies

Advances in ageing biology have opened novel therapeutic avenues that directly target mechanisms underlying age-related insulin resistance. Senotherapeutic strategies, including senolytics and senomorphics, aim to reduce the burden or pathological effects of senescent cells and their pro-inflammatory secretory phenotype (Saliev, Singh, 2025). Preclinical models and early-phase human trials demonstrate that clearance of senescent cells improves insulin sensitivity, physical function, and metabolic parameters (Xu et al., 2018; Justice et al., 2019). Nevertheless, clinical translation remains in its infancy, and long-term safety and efficacy data in older populations are still lacking.

Targeting mitochondrial health represents another promising frontier. Interventions aimed at enhancing mitochondrial biogenesis, improving mitophagy, or restoring organelle crosstalk—through exercise mimetics, NAD<sup>+</sup> precursors, or lifestyle-based approaches—have shown improvements in metabolic flexibility in early human studies (Short et al., 2005; Elhassan et al., 2019). However, robust randomized evidence specifically in frail or multimorbid older adults remains limited.

Brain insulin resistance has also emerged as an innovative therapeutic target (Chen et al., 2025). Proof-of-concept randomized trials of intranasal insulin demonstrate improvements in memory performance and modulation of brain metabolism in older adults and individuals at

risk for Alzheimer's disease, independent of peripheral glucose levels (Craft et al., 2012, 2017). While preliminary, these findings support the hypothesis that restoring insulin signaling in the brain may yield benefits extending beyond traditional metabolic endpoints.

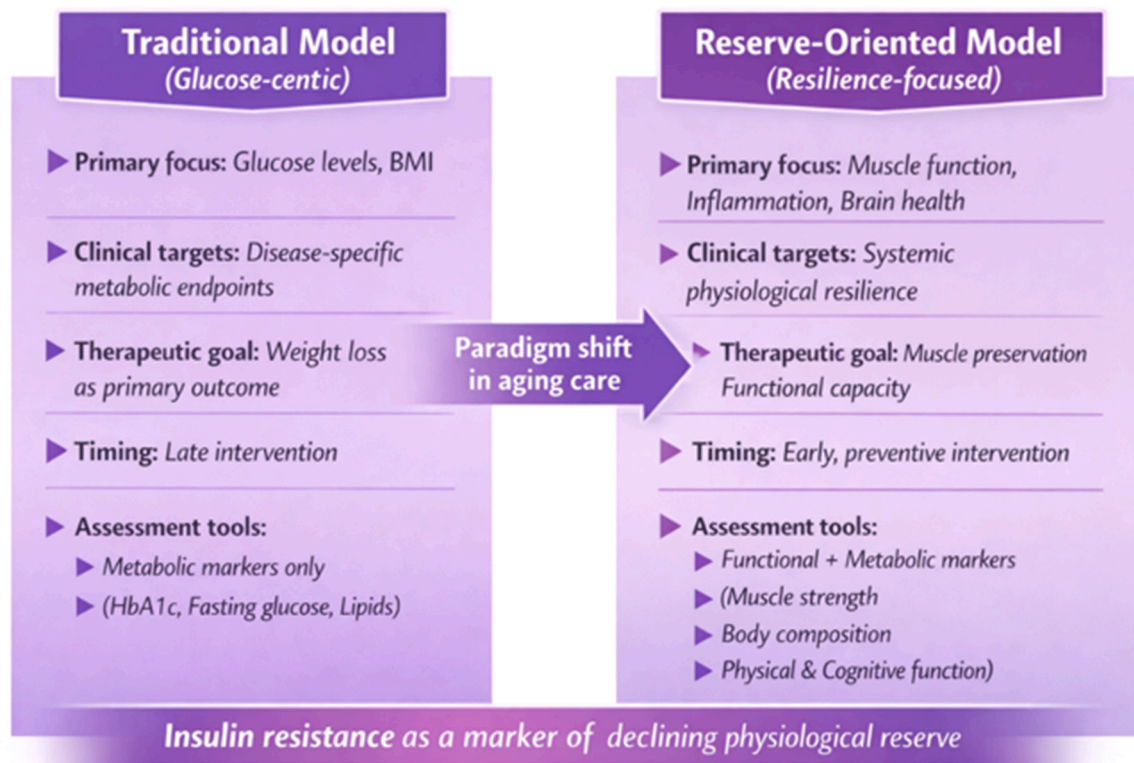
### 6.4. Toward an integrated, reserve-oriented clinical model

Taken together, these data support a paradigm shift from disease-specific management toward a reserve-oriented clinical model. In this framework, physiological reserve denotes the integrated capacity of metabolic, musculoskeletal, and neurocognitive systems to tolerate stressors, adapt to perturbations, and recover function. From this perspective, insulin resistance functions primarily as a biomarker of declining metabolic resilience rather than a discrete therapeutic target. Accordingly, interventions are evaluated not solely by their ability to lower glucose levels or reduce body weight, but by their capacity to preserve muscle function, limit inflammatory burden, and maintain cognitive and emotional health. This approach aligns closely with core geriatric principles that emphasize biological heterogeneity, functional status, and individualized risk–benefit assessment. It also provides a rationale for earlier intervention, before insulin resistance becomes entrenched and before progressive loss of physiological reserve translates into irreversible tissue damage. Screening strategies incorporating measures of muscle strength, body composition, and physical as well as cognitive performance may therefore offer greater clinical insight than traditional metabolic markers alone.

## 7. Conclusions

Existing reviews have largely addressed IR through an obesity/diabetes lens or as a peripheral metabolic defect. Here we synthesize evidence supporting IR in later life as a multisystem marker of declining metabolic resilience, integrating skeletal muscle, adipose tissue, and





**Fig. 2.** Conceptual shift from a glucose-centric to a reserve-oriented management model in older adults. The reserve-oriented approach prioritizes preservation of muscle function, reduction of inflammatory burden, and maintenance of cognitive health, integrating functional and metabolic markers to assess physiological resilience rather than focusing exclusively on glycaemic targets or weight reduction.

brain pathways and linking them to frailty, cognitive decline, and multimorbidity. Insulin resistance in ageing represents a convergence point of metabolic, inflammatory, and neurobiological processes that collectively undermine physiological resilience. Rather than viewing insulin resistance as a downstream consequence of obesity or an isolated precursor to diabetes, it should be recognized as a hallmark of metabolic ageing that integrates skeletal muscle decline, adipose tissue dysfunction, and brain vulnerability. This perspective explains the frequent co-occurrence of insulin resistance with frailty, cognitive impairment, and mood disorders in older adults and highlights the limitations of glucose-centric therapeutic paradigms (Fig. 1).

Future research should prioritize integrative strategies that address the biological drivers of metabolic ageing and focus on preserving functional reserve across organ systems. By aligning metabolic care with the principles of ageing biology and geriatric medicine, it may be possible to move beyond disease management toward the promotion of metabolic resilience and healthy ageing (Fig. 2).

#### Authors contribution

Conceptualization: V.B.; methodology, V.B.; investigation, V.B; A.J.S.; writing—review and editing, V.B.; supervision, A.J.S. All authors have read and agreed to the published version of the manuscript.

#### Informed consent statement

Not applicable.

#### Institutional review board statement

Not applicable.

#### Funding

This research received no external funding.

#### Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

#### Acknowledgments

None

#### Data availability

Not applicable.

#### References

- Abbatecola, A.M., Paolisso, G., 2008. Is there a relationship between insulin resistance and frailty syndrome? *Curr. Pharm. Des.* 14 (4), 405–410.
- Adler, A.I., Coleman, R.L., Leal, J., Whiteley, W.N., Clarke, P., Holman, R.R., 2024. Post-trial monitoring of a randomised controlled trial of intensive glycaemic control in type 2 diabetes extended from 10 years to 24 years (UKPDS 91). *Lancet* 404, 145–155. [https://doi.org/10.1016/S0140-6736\(24\)00537-3](https://doi.org/10.1016/S0140-6736(24)00537-3).
- Ahmed, B., Sultana, R., Greene, M.W., 2021. Adipose tissue and insulin resistance in obese. *Biomed. Pharm.* 137, 111315. <https://doi.org/10.1016/j.biopha.2021.111315>.
- Arnold, S.E., Arvanitakis, Z., et al., 2018. Brain insulin resistance in type 2 diabetes and Alzheimer disease. *Nat. Rev. Neurol.* 14 (3), 168–181.
- Arruda, A.P., Pers, B.M., Parlakg  l, G., G  ney, E., Inouye, K., Hotamisligil, G.S., 2014. Chronic enrichment of hepatic endoplasmic reticulum-mitochondria contact leads to mitochondrial dysfunction in obesity. *Nat. Med.* 20 (12), 1427–1435. <https://doi.org/10.1038/nm.3735>.





- Morley, J.E., 2022. Pharmacologic options for obesity and sarcopenic obesity in older persons. *Clin. Geriatr. Med.* 38 (1), 95–108.
- Mosenzon, O., Del Prato, S., Schechter, M., et al., 2021. From glucose lowering agents to disease/diabetes modifying drugs: a “SIMPLE” approach for the treatment of type 2 diabetes. *Cardiovasc. Diabetol.* 20, 92. <https://doi.org/10.1186/s12933-021-01281-y>.
- Myers, M.G., Heymsfield, S.B., Haft, C., et al., 2010. Leptin resistance. *Annu. Rev. Physiol.* 72, 537–556.
- Nowakowska-Lipiec, K., Zadoń, H., Michnik, R., Nawrat-Szoltysik, A., 2025. Progressive loss of muscle strength: the effects of ageing and sarcopenia on muscle function in older females. *J. Clin. Med.* 14 (20), 7276. <https://doi.org/10.3390/jcm14207276>.
- Palmer, A.K., Tchkonina, T., LeBrasseur, N.K., et al., 2019. Cellular senescence in adipose tissue. *J. Clin. Invest.* 129 (3), 966–981.
- Pedersen, B.K., Febbraio, M.A., 2012. Muscles, exercise and obesity: skeletal muscle as a secretory organ. *Nat. Rev. Endocrinol.* 8 (8), 457–465.
- Pellegrini, V., La Grotta, R., Carreras, F., et al., 2024. Inflammatory trajectory of type 2 diabetes: novel opportunities for early and late treatment. *Cells* 13, 1662. <https://doi.org/10.3390/cells13191662>.
- Perry, V.H., Holmes, C., 2014. Microglial priming in neurodegenerative disease. *Nat. Rev. Neurol.* 10 (4), 217–224. <https://doi.org/10.1038/nrneuro.2014.38>.
- Perseghin, G., Scifo, P., De Cobelli, F., et al., 1999. Intramyocellular triglyceride content is a determinant of in vivo insulin resistance. *Diabetes* 48 (8), 1600–1606.
- Petersen, K.F., Befroy, D., Dufour, S., et al., 2003. Mitochondrial dysfunction in the elderly: possible role in insulin resistance. *Science* 300 (5622), 1140–1142.
- Petersen, K.F., Dufour, S., Befroy, D., Garcia, R., Shulman, G.I., 2004. Impaired mitochondrial activity in insulin-resistant offspring of patients with type 2 diabetes. *N. Engl. J. Med.* 350 (7), 664–671.
- Petersen, M.C., Shulman, G.I., 2018. Mechanisms of insulin action and insulin resistance. *Physiol. Rev.* 98 (4), 2133–2223.
- Picard, M., McEwen, B.S., Epel, E.S., Sandi, C., 2018. An energetic view of stress: Focus on mitochondria. *Front. Neuroendocrinol.* 49, 72–85. <https://doi.org/10.1016/j.ynrne.2018.01.001>. PMID: PMC5964020.
- Popescu, C., Aurelian, S.M., Mirea, A., Munteanu, C., Vlădulescu-Trandafir, A.I., Anghelescu, A., Oancea, C., Andone, I., Spînu, A., Suci, A.V., et al., 2026. Sarcopenia as a multisystem disorder—connections with neural and cardiovascular systems: a related PRISMA systematic literature review. *Life* 16 (1), 68. <https://doi.org/10.3390/life16010068>.
- Porter, S.A., Massaro, J.M., Hoffmann, U., et al., 2023. Abdominal visceral and subcutaneous adipose tissue and metabolic risk in older adults. *J. Gerontol. A Biol. Sci. Med. Sci.*
- Prattichizzo, F., de Candia, P., De Nigris, V., Nicolucci, A., Ceriello, A., 2020. Legacy effect of intensive glucose control on major adverse cardiovascular outcome: systematic review and meta-analyses of trials according to different scenarios. *Metabolism* 110, 154308. <https://doi.org/10.1016/j.metabol.2020.154308>.
- Pujadas, G., De Nigris, V., Prattichizzo, F., et al., 2017. The dipeptidyl peptidase-4 inhibitor teneligliptin functions as antioxidant on human endothelial cells exposed to chronic hyperglycemia and metabolic high-glucose memory. *Endocrine* 56, 509–520. <https://doi.org/10.1007/s12020-016-1052-0>.
- Raut, S., Bhalerao, A., Powers, M., Gonzalez, M., Mancuso, S., Cucullo, L., 2023. Hypometabolism, Alzheimer's disease, and possible therapeutic targets: an overview. *Cells* 12 (16), 2019. <https://doi.org/10.3390/cells12162019>.
- Rodríguez-Mañas, L., Angulo, J., Carnicero, J.A., El Assar, M., García-García, F.J., Sinclair, A.J., 2022. Dual effects of insulin resistance on mortality and function in non-diabetic older adults: Findings from the Toledo Study of Healthy Aging. *Geroscience* 44, 1095–1108. <https://doi.org/10.1007/s11357-021-00384-4>.
- Rodríguez-Mañas, L., Laosa, O., Vellas, B., Paolisso, G., Topinkova, E., Oliva-Moreno, J., et al., 2019. European MID-Frail Consortium. Effectiveness of a multimodal intervention in functionally impaired older people with type 2 diabetes mellitus. *J. Cachex. Sarcopenia Muscle* 10 (4), 721–733. <https://doi.org/10.1002/jcsm.12432>.
- Saliev, T., Singh, P.B., 2025. Targeting senescence: a review of senolytics and senomorphics in anti-ageing interventions. *Biomolecules* 15, 860. <https://doi.org/10.3390/biom15060860>.
- Scalabrini, M., Turco, E., Davigo, I., Filadi, R., Nogara, L., Gherardi, G., Barazzuol, L., Armani, A., Trani, G., Negro, S., Franco-Romero, A., Jaspers, Y., Baschiera, E., De Cegli, R., Del Prete, E., Cali, T., Blaauw, B., Salviati, L., Rigoni, M., Mammucari, C., Caspar-Bauguil, S., Moro, C., Pizzo, P., Sandri, M., Kemp, S., Romanello, V., 2025. Alterations in peroxisome-mitochondria interplay in skeletal muscle accelerate muscle dysfunction. *Nat. Commun.* 16 (1), 9868. <https://doi.org/10.1038/s41467-025-64833-w>.
- Short, K.R., Bigelow, M.L., Kahl, J., et al., 2005. Decline in skeletal muscle mitochondrial function with ageing in humans. *Proc. Natl. Acad. Sci. USA* 102 (15), 5618–5623. <https://doi.org/10.1073/pnas.0501559102>.
- Short, K.R., Vittone, J.L., Bigelow, M.L., et al., 2003. Impact of aerobic exercise training on age-related changes in insulin sensitivity and muscle oxidative capacity. *Diabetes* 52 (8), 1888–1896.
- Shulman, G.I., 2000. Cellular mechanisms of insulin resistance. *J. Clin. Invest.* 106 (2), 171–176.
- Sinclair, A., Siqueira, I., Abdelhafiz, A., 2025. The effect of frailty on body composition and its impact on the use of SGLT-2 inhibitors and GLP-1 receptor agonists in older persons with diabetes. *Metabolites* 15 (6), 381. <https://doi.org/10.3390/metabo15060381>.
- Stephens, N.A., Sparks, L.M., 2007. Resistance training improves insulin sensitivity independent of weight loss. *Appl. Physiol. Nutr. Metab.* 32 (5), 963–971. <https://doi.org/10.1139/H07-093>.
- Stout, M.B., Justice, J.N., Nicklas, B.J., Kirkland, J.L., 2017. Physiological ageing: links among adipose tissue dysfunction, diabetes, and frailty. *Physiology* 32 (1), 9–19.
- Tchkonina, T., Morbeck, D.E., von Zglinicki, T., et al., 2010. Fat tissue, ageing, and cellular senescence. *Ageing Cell* 9 (5), 667–684.
- Tero-Vescan, A., Degens, H., Matsakas, A., Ștefănescu, R., Ösz, B.E., Slevin, M., 2025. Exercise-induced muscle-fat crosstalk: molecular mediators and their pharmacological modulation for the maintenance of metabolic flexibility in ageing. *Pharmaceuticals* 18 (8), 1222. <https://doi.org/10.3390/ph18081222>.
- Veldhuis, J.D., Roelfsema, F., Keenan, D.M., Pincus, S.M., 2011. Endocrine mechanisms of ageing. *Endocr. Rev.* 32 (4), 393–448.
- Villareal, D.T., Chode, S., Parimi, N., et al., 2011. Weight loss, exercise, or both and physical function in obese older adults. *N. Engl. J. Med.* 364 (13), 1218–1229. <https://doi.org/10.1056/NEJMoa1008234>.
- Virtue, S., Vidal-Puig, A., 2010. Adipose tissue expandability, lipotoxicity and metabolic disease. *Nat. Rev. Endocrinol.* 6, 429–438.
- Volpi, E., Mittendorfer, B., Rasmussen, B.B., Wolfe, R.R., 2000. The response of muscle protein anabolism to amino acids is impaired in the elderly. *J. Clin. Invest.* 106 (2), 201–209. <https://doi.org/10.1172/JCI10570>.
- Volpi, E., Rasmussen, B.B., et al., 2018. Muscle tissue changes with ageing. *Curr. Opin. Clin. Nutr. Metab. Care* 21 (1), 19–25.
- Weavers, H., 2024. Biological resilience in health and disease. *Dis. Model Mech.* 17 (7), dmm050799. <https://doi.org/10.1242/dmm.050799>.
- Willette, A.A., Xu, G., Johnson, S.C., Birdsill, A.C., Jonaitis, E.M., Sager, M.A., Hermann, B.P., La Rue, A., Asthana, S., Bendlin, B.B., 2013. Insulin resistance, brain atrophy, and cognitive performance in late middle-aged adults. *Diabetes Care* 36 (2), 443–449. <https://doi.org/10.2337/dc12-0922>.
- Xu, H., Barnes, G.T., Yang, Q., et al., 2015. Chronic inflammation in fat plays a crucial role in the development of insulin resistance. *J. Clin. Invest.* 125, 1821–1830.
- Xu, M., Pirtskhalava, T., Farr, J.N., et al., 2018. Senolytics improve physical function and increase lifespan in old age. *Nat. Med.* 24 (8), 1246–1256. <https://doi.org/10.1038/s41591-018-0092-9>.
- Zamboni, M., Mazzali, G., Fantin, F., Rossi, A., Di Francesco, V., 2008. Sarcopenic obesity. *Am. J. Clin. Nutr.* 87, 801–806.
- Zhang, X., Gao, Y., Zhang, S., Wang, Y., Pei, X., Chen, Y., Zhang, J., Zhang, Y., Du, Y., Hao, S., Wang, Y., Ni, T., 2025. Mitochondrial dysfunction in the regulation of ageing and ageing-related diseases. *Cell Commun. Signal* 23 (1), 290. <https://doi.org/10.1186/s12964-025-02308-7>.